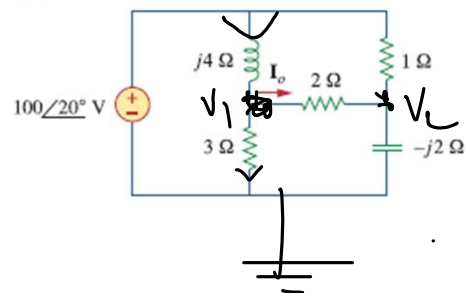


10.17 By nodal analysis, obtain current I_o in the circuit of

Fig. 10.66.



$$\frac{100\angle 20^\circ - V_1}{j4} - \frac{V_1 - V_2}{2} - \frac{V_1}{3} = 0$$

$$25\angle 110^\circ$$

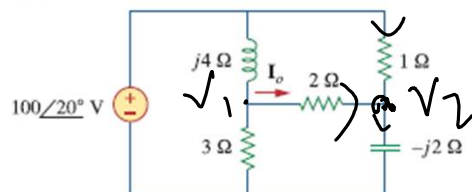
$$8.87\angle 163.3^\circ$$

$\Rightarrow$

$$V_1 \left(\frac{-1}{j4} - \frac{1}{2} - \frac{1}{3} \right) + \frac{V_2}{2} = -25\angle -70^\circ$$

10.17 By nodal analysis, obtain current I_o in the circuit of

Fig. 10.66.



$$1.58 \angle -161.6^\circ$$

$$KCL - 1$$

$$\frac{100 \angle 20^\circ - V_2}{1} + \frac{V_1 - V_2}{2}$$

$$-\frac{V_2}{-j2} = 0$$

$$\Rightarrow \frac{V_1}{2} + V_2 \left(-1 - \frac{1}{2} + \frac{1}{j2} \right) = 100 \angle 20^\circ$$

$$\Delta \Rightarrow \begin{bmatrix} 0.83 \angle 162.3^\circ & \frac{1}{2} \\ \frac{1}{2} & 1.58 \angle 161.6^\circ \end{bmatrix}$$

$$V_1 \rightarrow \Delta_1 = \begin{bmatrix} 25 \angle 110^\circ & \frac{1}{2} \\ 100 \angle 20^\circ & 1.58 \angle 161.6^\circ \end{bmatrix}$$

$$V_2 \rightarrow \Delta_2 = \begin{bmatrix} 0.83 \angle 163.3^\circ \\ \frac{1}{2} \end{bmatrix}$$

$25 \angle 110^\circ$
 $100 \angle 20^\circ$

$$\Delta = 1.12 \angle -42.5^\circ$$

$$\Delta_1 = 72.85 \angle -129^\circ$$

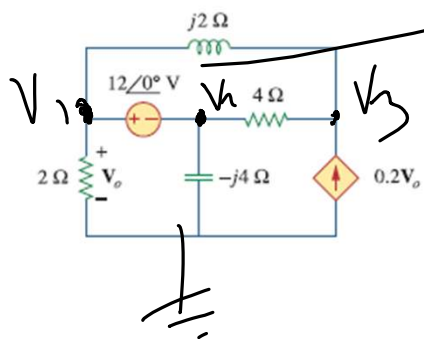
$$\Delta_2 = 80.31 \angle -168.1^\circ$$

$$V_1 = \frac{\triangle}{\triangle}$$

$$V_2 = \frac{\triangle 2}{\triangle}$$

$$I_0 = \frac{V_2 - V_1}{2}$$

10.19 Obtain V_o in Fig. 10.68 using nodal analysis.



super node.

super node eqⁿ

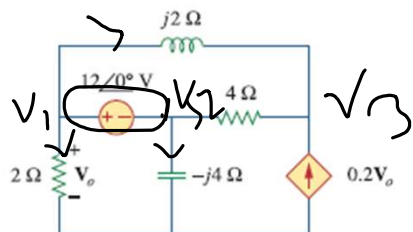
$$V_1 - V_2 = 12 \angle 0^\circ$$

$$V_o = V_1$$

$$V_1 = 12 \angle 0^\circ + V_2$$

$$V_o = V_{2\Omega} = V_1$$

10.19 Obtain V_o in Fig. 10.68 using nodal analysis.



KCL - at super node

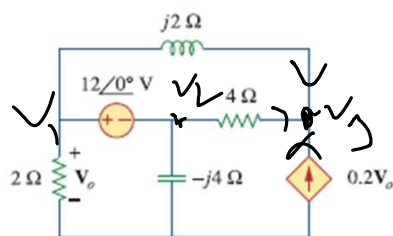
$$\frac{V_1}{2} + \frac{V_2}{-j4} + \frac{V_1 - V_3}{j2} + \frac{V_2 - V_3}{4} = 0$$

$$\Rightarrow V_1 \left(\frac{1}{2} + \frac{1}{j2} \right) + V_2 \left(\frac{1}{-j4} + \frac{1}{4} \right) + V_3 \left(-\frac{1}{j2} - \frac{1}{4} \right) = 0$$

3x3

ucll-2

10.19 Obtain V_o in Fig. 10.68 using nodal analysis.



$$0.2 \times V_o = 0.2 \times V_1$$

$$\frac{V_1 - V_3}{j2} + \frac{V_2 - V_3}{4} + 0.2V_1 = 0$$

$$\Rightarrow V_1 \left(\frac{1}{j2} + 0.2 \right) + V_2 \times \frac{1}{4} + V_3 \left(\frac{-1}{j2} - \frac{1}{4} \right) = 0$$

Substituting $V_1 = 12L\ddot{\theta} + V_2$

$$\begin{aligned} (12L\ddot{\theta} + V_2) \left(\frac{1}{2} + \frac{1}{j^2} \right) + V_2 \left(\frac{1}{-j^4} + \frac{1}{4} \right) \\ + V_3 - \frac{1}{2} - \frac{1}{4} \\ = 0 \end{aligned}$$

Substituting $V_1 = 12\ddot{\theta} + V_2$

$$\left(12\ddot{\theta} + V_2 \right) \left(\frac{1}{j^2} + 0.2 \right) + V_2 \times \frac{1}{4} + V_3 \left(\frac{-1}{j^2} - \frac{1}{4} \right) = 0$$

AC ckt Power

DC $\rightarrow$

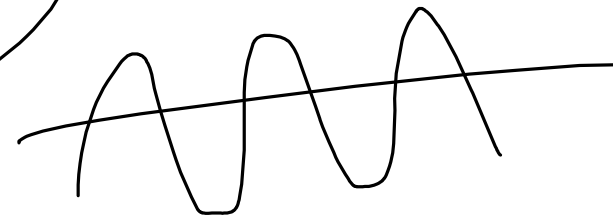
R

C

I

$$P = VI$$

o o



$$Z_L = X_L \overset{\omega L}{\underset{\omega L}{\curvearrowright}} \overset{\omega L}{\underset{\omega L}{\curvearrowright}} \rightarrow X_L j$$

$$Z_C = X_C \overset{\omega C}{\underset{\omega C}{\curvearrowright}} \rightarrow X_C^{-j}$$

AC Ckt Power

$$Z = R + jX$$

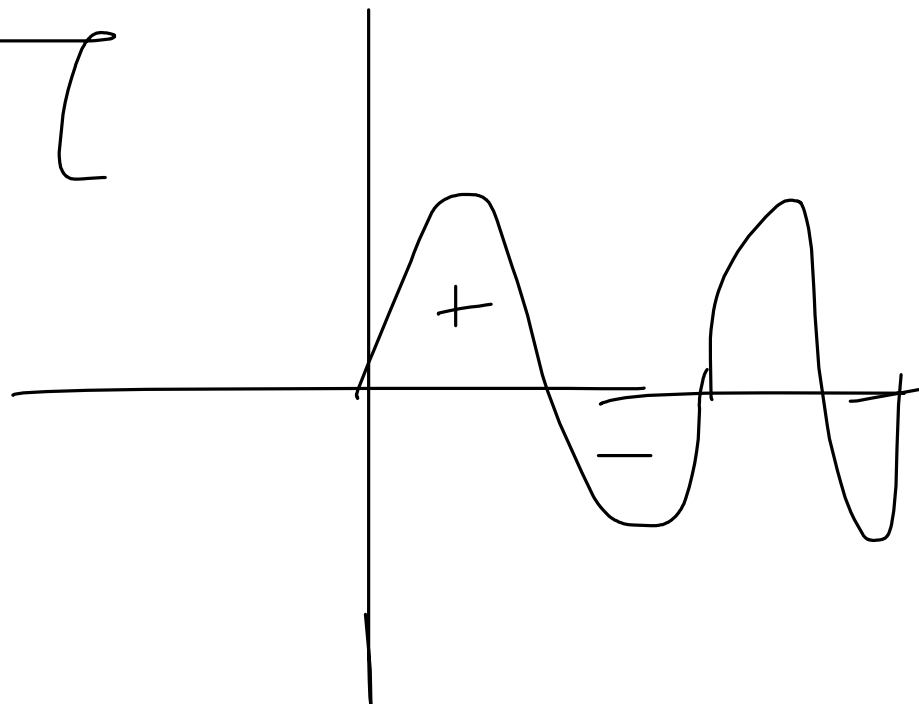
- ✓ ① Real power $\rightarrow R$
- ② Reactive " $\rightarrow L, C$
- ③ Complex " $\rightarrow Z$
- ④ Apparent " $\rightarrow Z$

$$X = X_C - X_L$$

C, L



τ



$V_L \theta V$



$C \rightarrow$ stores energy
in E-field

$L \rightarrow$ store energy in
Magnetic field.

$$\text{comp } I_e \times = S \angle \theta_S$$

$$\text{apparent power} = |S|$$

$$\boxed{I^2 R}$$

$$Z = R + jX$$

Real power $|P| = |I|^2 R$

Reactive power, $|Q| = |I|^2 X$

Complex power, $\bar{S} = |I|^2 Z$

Apparent power,

$$= |\bar{S}|$$

$$= |I|^2 \times |Z|$$